

Problem 1 (§7, #8). Suppose $a, b \in \mathbb{Z}$. Prove that $a \equiv b \pmod{10}$ if and only if $a \equiv b \pmod{2}$ and $a \equiv b \pmod{5}$.

Problem 2 (§7, #14). Suppose $a \in \mathbb{Z}$. Prove that $a^2 \mid a$ if and only if $a \in \{-1, 0, 1\}$.

Problem 3 (§7, #14). If $\gcd(a, c) = \gcd(b, c) = 1$, then $\gcd(ab, c) = 1$.

Problem 4 (§8, #4). If $m, n \in \mathbb{Z}$, then $\{x \in \mathbb{Z} : mn \mid x\} \subseteq \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$.

Problem 5 (§8, #18). If A, B and C are sets, then $A \times (B - C) = (A \times B) - (A \times C)$.

Problem 6 (§8, #28). Prove that $\{12a + 25b : a, b \in \mathbb{Z}\} = \mathbb{Z}$.

Problem 7 (§9, #30). Disprove the statement by proving its negation.

There exist integers a and b for which $42a + 7b = 1$.

Problem 8 (§9, #21 (adapted)). Disprove the statement by contradiction.

There exist prime numbers p and q for which $p - q = 2023$.